

Job Sequencing and Tool Switching

Structural results and a branch-and-Benders-cut study

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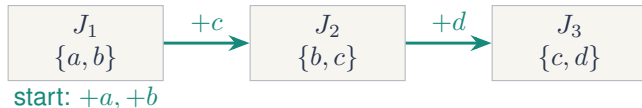
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The order decides which tools return

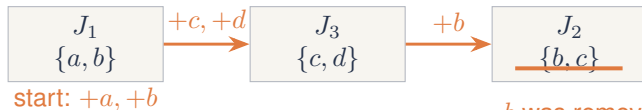
Magazine capacity $b = 2$; the machine starts empty.

order 1, 2, 3



4 insertions

order 1, 3, 2



5 insertions

b was removed, then returned

Report: Figures 1–2 and Section 1.1.

Loading is easy; the order is the hard part

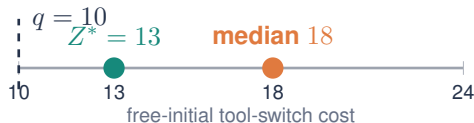
Fixed order

When a tool must leave, remove the loaded tool whose next use is furthest away.

$$\text{KTNS}(\pi) \text{ in } O(n|U|).$$

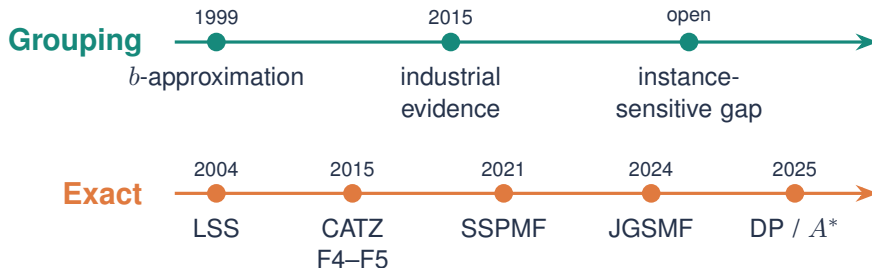
Choosing the order is NP-hard for every fixed $b \geq 2$.

40,320
orders of the eight-job instance L1-1
180
are optimal



Tang-Denardo KTNS; Crama et al. hardness; report Figure 3.

Two gaps the literature leaves open

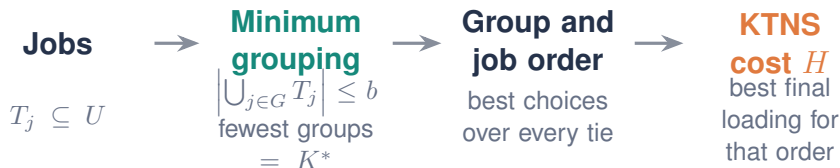


Question 1 What does minimising the number of groups cost after KTNS?

Question 2 Can an exact loading oracle make Benders competitive with compact models?

Report: literature review in Chapter 3.

Grouping commits before it sees the sequence



$$Z^*$$

best KTNS cost over every job order

$$H - Z^*$$

loss that remains even under favourable
tie-breaking

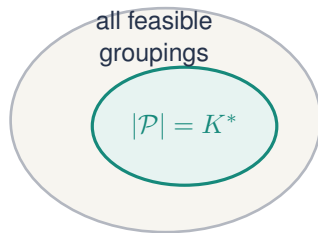
Report: Definitions 5–7.

Drop the fewest-groups rule: grouping is exact

$$Z^* = \min_{\mathcal{P} \text{ feasible}} \gamma(\mathcal{P})$$

$$H_{\text{walk}} = \min_{\substack{\mathcal{P} \text{ feasible} \\ |\mathcal{P}|=K^*}} \gamma(\mathcal{P}), \quad H \leq H_{\text{walk}}.$$

The approximation enters when the first phase keeps only the groupings with the fewest groups.



$$\text{cost} = q + R$$

$q = |U| - b$; R counts reinsertions.

Report: Theorem 1, Corollary 1 and Lemma 1.

Four jobs are enough for a genuine grouping loss

Required tools; capacity $b = 4$

	1	2	3	4	5	6	7
A	●	●					
B			●	●			
C	●		●		●	●	
D		●			●	●	●

A and B are the only feasible pair.

Unrestricted order

$$B, C, A, D \implies Z^* = q = 3.$$

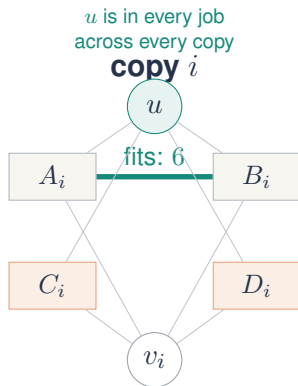
Every minimum grouping

$$A \text{ and } B \text{ consecutive} \implies H = 4.$$

One forced adjacency creates one reinsertion.

Report: Proposition 10.

One shared tool makes the family connected



- ▶ C_i and D_i fill all six slots.
- ▶ $A_i \cup B_i$ uses exactly six tools.
- ▶ Small jobs from different copies use seven.

**unique minimum
grouping**

$$K^* = 3g$$

Report: Theorem 5, construction and first proof step.

The unrestricted order reaches coverage



u remains loaded

repeat copy by copy; every newly loaded private tool is appearing for the first time

$$|U| = 8g + 1, \quad b = 6$$

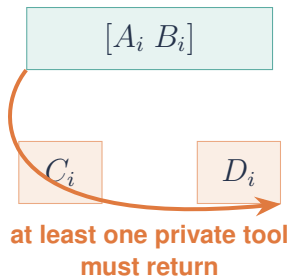
$$q = |U| - b = 8g - 5.$$

No reinsertion is needed, so the coverage lower bound is attained:

$$Z^* = q = 8g - 5.$$

Report: Theorem 5, coverage schedule.

Every minimum grouping forces one return per copy



Lower bound

$$H \geq q + g = (8g + 1 - 6) + g = 9g - 5.$$

Matching grouped order

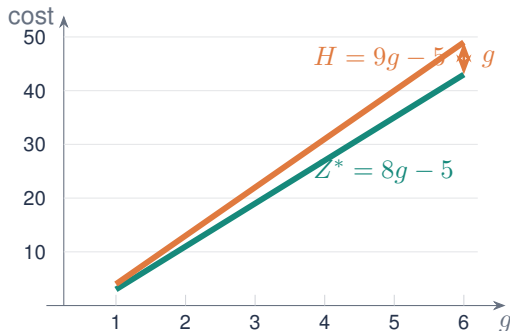
$$A_i, B_i, C_i, D_i$$

has exactly one return per copy.
Therefore

$$H = 9g - 5.$$

Report: Theorem 5, forced-return count and matching schedule.

The additive gap grows without bound



$$H - Z^* = g$$

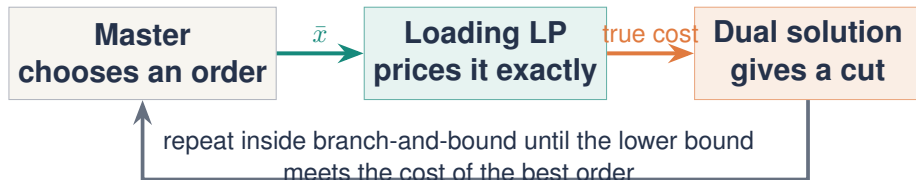
The additive loss is unbounded, even though all jobs remain connected through u .

$$\frac{H}{Z^*} \longrightarrow \frac{9}{8}.$$

This family does not prove a universal $9/8$ ratio bound.

Report: connected unbounded heuristic-gap theorem.

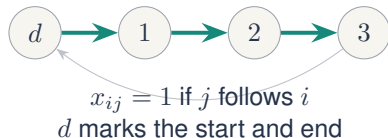
Order in the master, loading in an exact oracle



The decomposition is natural because the fixed-order subproblem is polynomial and exact.

Report: Figure 9 and Section 5.2.

The master starts from local and coverage bounds



$$\begin{aligned} \min \theta, \quad & \sum_{i \neq j} x_{ij} = 1, \quad \sum_{j \neq i} x_{ij} = 1, \\ & \sum_{i,j \in S} x_{ij} \leq |S| - 1 \quad (S \subseteq J). \end{aligned}$$

Starting lower bounds

$$\theta \geq |U|, \quad \theta \geq \sum_{i,j \in J} w_{ij} x_{ij} + \sum_{j \in J} |T_j| x_{dj},$$

where $w_{ij} = \max(0, |T_i \cup T_j| - b)$.

Report: Benders master formulation, Section 5.2.

The loading LP is integral on a fixed path

$$\begin{aligned} \min \quad & \sum_{j,t} z_{jt} \\ \text{s.t.} \quad & y_{jt} \geq 1 & (t \in T_j), \\ & \sum_t y_{jt} \leq b & (j \in J), \\ & z_{jt} \geq y_{jt} - y_{it} - (1 - x_{ij}) & (i \neq j), \\ & z_{jt} \geq y_{jt} - (1 - x_{dj}) & (j \in J), \\ & y, z \geq 0. \end{aligned}$$

y_{jt} records whether tool t is loaded at job j .

z_{jt} charges an insertion when the selected path moves into j .

fixed path
↓
exact KTNS cost

Report: fixed-order loading formulation and integrality argument.

The dual cut is supported only on arcs

$$\theta \geq \alpha + \sum_{i \neq j} \beta_{ij} x_{ij}, \quad \beta_{ij} = \sum_t \bar{\lambda}_{ijt}.$$

Valid globally

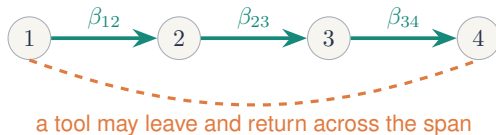
The dual feasible region does not depend on the proposed order.

Tight locally

An optimal dual prices the generating order exactly.

Arc-supported

Every learned coefficient sits on one predecessor arc.



Report: Equation 20 and locality discussion in Chapter 7.

The campaign tests 12 solver configurations

Compact baselines

SSPMF

CATZ-F4

LSS

Reimplementations used for a controlled comparison, not a literature-wide ranking.

Benders regimes

BBC-LP

BBC-LP+T

BBC-K

BBC-LP+F

BBC-K+F

BBC-LP+F+H

BBC-LP+F+C

BBC-LP+F+P

BBC-LP+ACC

1,410 canonical benchmark instances

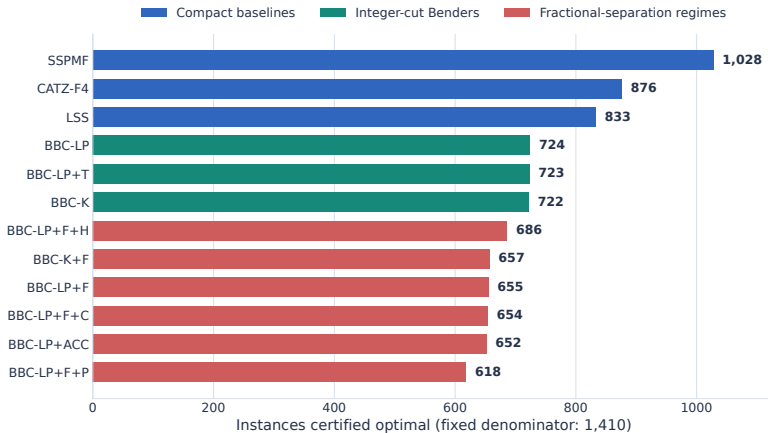
12 configurations

16,920 canonical outcomes

1 **hour** per run; fixed denominator

Report: experimental design and Table 5.

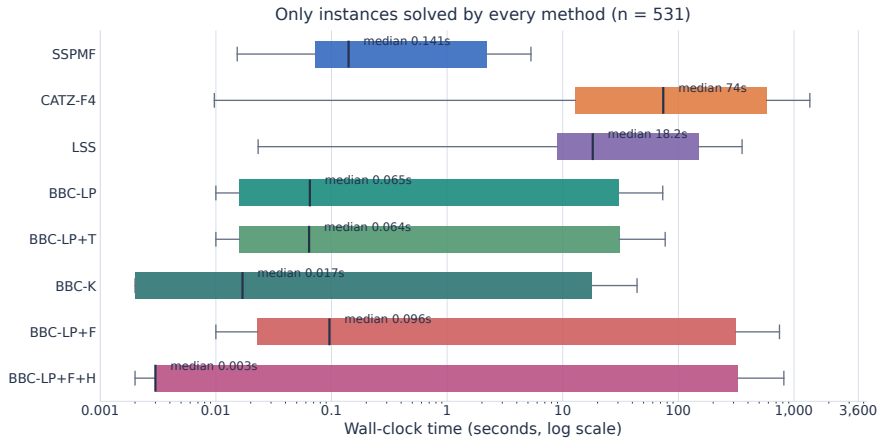
Every compact baseline certifies more than Benders



Each bar uses the same 1,410-instance denominator.

Canonical 16,920-row campaign; report Table 6.

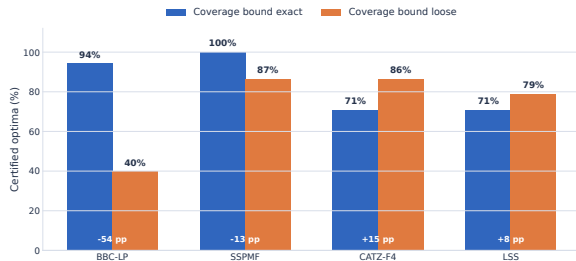
Benders is quick, with a long tail



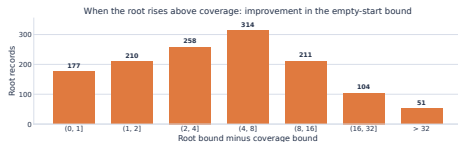
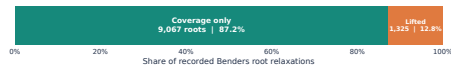
The same 531 instances appear in every box; the horizontal axis is logarithmic.

Canonical campaign; common-solved comparison in report Table 7.

Closure drops when the bound misses the optimum



1,107 instances with a known optimum.



223/383 time-limited, known-optimum BBC-LP incumbents already equal the optimum.

Report: Table 10, Observation 13 and the incumbent audit.

Next: pricing tool residency across a span

What the theory isolates

Minimum grouping can force a tool out and back:

$$\text{cost} = q + \text{reinsertions}.$$

What the campaign isolates

The tested master often finds the right order but cannot certify those returns.

The work narrows the next experiment; it claims no new SOTA solver.

Next exact tests

- ▶ disaggregate θ by position;
- ▶ encode job-union structure over a window;
- ▶ test the stronger bound under a matched protocol.

Use the grouping result

- ▶ search $K^*, \dots, K^* + \delta$ groupings;
- ▶ flatten each with KTNS; use the orders as seeds.

Report: Discussion and further work, Chapters 7–8.

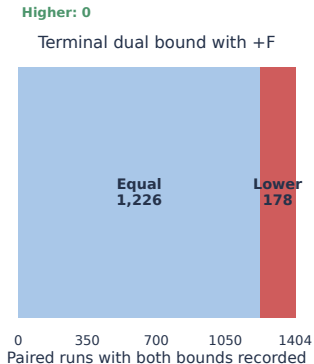
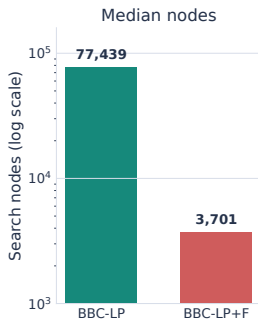
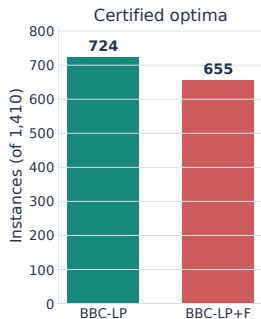
Backup: certified optima by method and family

Configuration	Cat.	Crama	Lap.3	Lap.4	Lap.5	Lap.7	All
SSPMF	62	68	340	330	162	66	1028
CATZ-F4	44	44	340	330	74	44	876
LSS	41	42	340	330	39	41	833
BBC-LP+T	34	34	340	192	84	39	723
BBC-LP	34	35	340	192	84	39	724
BBC-K	35	35	340	190	83	39	722
BBC-LP+F+H	39	37	316	157	97	40	686
BBC-LP+F	33	32	313	156	85	36	655
BBC-LP+F+C	33	32	311	157	85	36	654
BBC-LP+ACC	39	37	286	154	97	39	652
BBC-K+F	33	31	316	156	86	35	657
BBC-LP+F+P	31	32	282	155	83	35	618

Family denominators: 160, 160, 340, 330, 340, 80; total 1,410.

Report: Table 6.

Backup: fractional cuts spend more, certify less



BBC-LP+F generated 67.0 million fractional cuts. The +F regime also disables presolve.

Against BBC-LP+F: $+H$ certifies 31 more; $+C$ certifies 1 fewer; $+P$ certifies 37 fewer; all three together certify 3 fewer.

Report: Tables 8–9.

Backup: proof and validation ledger

Connected-family proof

$K^* = 3g$ unique feasible pair in each copy

$Z^* = 8g - 5$ coverage bound plus a schedule with no return

$H = 9g - 5$ one forced return per copy plus a matching order

connected universal tool u lies in every job

Independent checks

349,872 fixed-order loading checks against exact dynamic programming

988 instances proved optimal by at least two methods

41,514 pairwise method comparisons, with no objective disagreement

16,920 canonical campaign outcomes audited

Formal scope and remaining conjectures are recorded separately in `THEOREM_AUDIT.md`.

Report: Appendices A–C and the independent verification program.